

Practical - 9

Aim:

Connect Python to the MySQL database.

Q1: Write a python code to establish connection with MySQL and create database demo_data. Also display list of all the available database.

Code:

```
import mysql.connector

config = {
    'user' : 'root',
    'password' : 'vatsal',
    'host' : 'localhost',
}

try:
    cnx = mysql.connector.connect(**config)
    if cnx.is_connected():
        print("Connected to MySQL database")
except mysql.connector.Error as err:
    print(f"Error: {err}")

cursor = cnx.cursor()

query = "CREATE DATABASE IF NOT EXISTS demo_data"
cursor.execute(query)
cnx.commit()
query = "SHOW DATABASES"
cursor.execute(query)
results = cursor.fetchall()
print("Databases:")
for row in results:
    print(f"Name: {row[0]}")

query = "USE demo_data"
cursor.execute(query)
```

Output:

```
Connected to MySQL database
Databases:
Name: demo_data
Name: mysql
```

Q2: Write a python code to create below specified two tables 'hospital_details' and 'doctor_details' and insert values as mentioned below.

hospital_details		
Hospital_Id	Hospital_Name	Bed_count
1	Janta	200
2	Zydus	500
3	Sal	1000
4	Stirling	1500

doctor_details					
Doctor_Id	Doctor_Name	Hospital_Id	Speciality	Salary	Experience
101	Karan	1	Pediatric	40000	0
102	Naresh	1	Onchologist	80000	5
103	Hardik	2	Surgen	60000	2
104	Vishal	2	Homeopathy	50000	1
105	Jay	3	Aayurvedic	40000	0
106	Deep	3	Physeotherapist	70000	4
107	Divyesh	4	Pediatric	55000	3
108	Arjun	4	Scin	55000	3

Create TABLE 1:

```

query = """
CREATE TABLE IF NOT EXISTS hospital_details (
    id INT AUTO_INCREMENT PRIMARY KEY,
    hospital_name VARCHAR(255),
    bed_count INT
)
"""

cursor.execute(query)
print("Table 'hospital_details' created successfully")

insert_query = "INSERT INTO hospital_details (hospital_name, bed_count) VALUES (%s, %s)"
data = [
    ("Janta", 200),
    ("Zydus", 500),
    ("Sal", 1000),

```

```

        ("Stirling", 1500),
    ]
    cursor.executemany(insert_query, data)
    cnx.commit()
    print(f"{cursor.rowcount} records inserted successfully")

```

Output:

```

Table 'hospital_details' created successfully
4 records inserted successfully

```

Create TABLE 2:

```

query = """
CREATE TABLE IF NOT EXISTS doctor_details (
    id INT AUTO_INCREMENT PRIMARY KEY,
    doctor_name VARCHAR(255),
    hospital_id INT,
    speciality VARCHAR(255),
    salary DECIMAL(10, 2),
    experience INT
)
"""

cursor.execute(query)
print("Table 'doctor_details' created successfully")

insert_query = "INSERT INTO doctor_details (doctor_name, hospital_id, speciality,
salary, experience) VALUES (%s, %s, %s, %s, %s)"
data = [
    ("Karan", 1, "Pediatric", 40000.00, 0),
    ("Naresh", 1, "Oncology", 80000.00, 5),
    ("Hardik", 2, "Surgery", 60000.00, 2),
    ("Vishal", 2, "Homeopathy", 50000.00, 1),
    ("Jay", 3, "Ayurvedic", 40000.00, 0),
    ("Deep", 3, "Physiotherapist", 70000.00, 4),
    ("Divyesh", 4, "Pediatric", 55000.00, 3),
    ("Arjun", 4, "Scin", 55000.00, 3),
]
cursor.executemany(insert_query, data)
cnx.commit()
print(f"{cursor.rowcount} records inserted successfully")

```

Output:

```

Table 'doctor_details' created successfully
8 records inserted successfully

```

Q3: Write a python code to retrieve all the details of doctors.

Code:

```
query = "SELECT * FROM doctor_details"
cursor.execute(query)
results = cursor.fetchall()
print("Doctor Details:")
for row in results:
    print(f"ID: {row[0]}, Name: {row[1]}, Specialization: {row[3]}, Salary: {row[4]},
Experience: {row[5]}")
```

Output:

```
Doctor Details:
ID: 1, Name: Karan, Specialization: Pediatric, Salary: 40000.00,
Experience: 0
ID: 2, Name: Naresh, Specialization: Oncology, Salary: 80000.00,
Experience: 5
ID: 3, Name: Hardik, Specialization: Surgery, Salary: 60000.00,
Experience: 2
ID: 4, Name: Vishal, Specialization: Homeopathy, Salary: 50000.00,
Experience: 1
ID: 5, Name: Jay, Specialization: Ayurvedic, Salary: 40000.00,
Experience: 0
ID: 6, Name: Deep, Specialization: Physiotherapist, Salary: 70000.00,
Experience: 4
ID: 7, Name: Divyesh, Specialization: Pediatric, Salary: 55000.00,
Experience: 3
ID: 8, Name: Arjun, Specialization: Scin, Salary: 55000.00, Experience: 3
```

Q4: Write a python code to retrieve all the doctors who are in Janta hospital.

Code:

```
query = "SELECT d.id, d.doctor_name, d.speciality FROM doctor_details d JOIN
hospital_details h ON d.hospital_id = h.id WHERE h.hospital_name = 'Janta'"
cursor.execute(query)
results = cursor.fetchall()
print("Doctor Details:")
for row in results:
    print(f"ID: {row[0]}, Name: {row[1]}, Specialization: {row[2]}")
```

Output:

```
Doctor Details:
ID: 1, Name: Karan, Specialization: Pediatric
ID: 2, Name: Naresh, Specialization: Oncology
```

Q5: Write a python code to update experience of doctors.

Code:

```
query = "UPDATE doctor_details SET EXPERIENCE = EXPERIENCE + 1"
cursor.execute(query)
cnx.commit()
query = "SELECT * FROM doctor_details"
cursor.execute(query)
results = cursor.fetchall()
print("Doctor Details:")
for row in results:
    print(f"ID: {row[0]}, Name: {row[1]}, Experience: {row[5]}")
```

Output:

```
Doctor Details:
ID: 1, Name: Karan, Experience: 2
ID: 2, Name: Naresh, Experience: 7
ID: 3, Name: Hardik, Experience: 4
ID: 4, Name: Vishal, Experience: 3
ID: 5, Name: Jay, Experience: 2
ID: 6, Name: Deep, Experience: 6
ID: 7, Name: Divyesh, Experience: 5
ID: 8, Name: Arjun, Experience: 5
```

Q6: Write a python code to drop table 'hospital_details'.

Code:

```
print("Before Deletion:")
query = "SHOW TABLES"
cursor.execute(query)
results = cursor.fetchall()
for row in results:
    print(f"Name: {row[0]}")

query = "DROP TABLE hospital_details"
cursor.execute(query)
cnx.commit()

print("After Deletion:")
query = "SHOW TABLES"
cursor.execute(query)
results = cursor.fetchall()
for row in results:
    print(f"Name: {row[0]}")
```

Output:

```
Before Deletion:
Name: doctor_details
Name: hospital_details
After Deletion:
Name: doctor_details
```

Q7: Write a python code to create in memory database and table using sqlite database engine.

Code:

```
import sqlite3

cnx = sqlite3.connect(':memory:')
cursor = cnx.cursor()

query = "DROP TABLE IF EXISTS student_details"
cursor.execute(query)
cnx.commit()

query = """
CREATE TABLE IF NOT EXISTS student_details (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    student_name TEXT,
    age INTEGER,
    course TEXT
)
"""
cursor.execute(query)

query = """
INSERT INTO student_details (student_name, age, course) VALUES (?, ?, ?)
"""
data = [
    ("Vatsal", 21, "Computer Engineering"),
    ("Honey", 22, "Information Technology"),
    ("Aryan", 20, "MBBS"),
]
cursor.executemany(query, data)
cnx.commit()

query = "SELECT * FROM student_details"
cursor.execute(query)
results = cursor.fetchall()
print("Student Details:")
for row in results:
    print(f"ID: {row[0]}, Name: {row[1]}, Age: {row[2]}, Course: {row[3]}")
```

Output:

Student Details:

```
ID: 1, Name: Vatsal, Age: 21, Course: Computer Engineering
ID: 2, Name: Honey, Age: 22, Course: Information Technology
ID: 3, Name: Aryan, Age: 20, Course: MBBS
```

Q8: Establish connection with sqlite database engine and create above tables into database demo. Show demonstration of execute script to execute multiple queries at a time.

Code:

```
import sqlite3

cnx = sqlite3.connect(':memory:')
cursor = cnx.cursor()

query = "DROP TABLE IF EXISTS doctor_details"
cursor.execute(query)
cnx.commit()

query = "DROP TABLE IF EXISTS hospital_details"
cursor.execute(query)
cnx.commit()

query = """
CREATE TABLE IF NOT EXISTS hospital_details (
    id INT AUTO_INCREMENT PRIMARY KEY,
    hospital_name VARCHAR(255),
    bed_count INT
)
"""
cursor.execute(query)
print("Table 'hospital_details' created successfully")

insert_query = "INSERT INTO hospital_details (hospital_name, bed_count) VALUES (?, ?)"
data = [
    ("Janta", 200),
    ("Zydus", 500),
    ("Sal", 1000),
    ("Stirling", 1500),
]
cursor.executemany(insert_query, data)
cnx.commit()
print(f"{cursor.rowcount} records inserted successfully")

query = "SELECT * FROM hospital_details"
cursor.execute(query)
results = cursor.fetchall()
print("\nHospital Details:")
for row in results:
    print(f"ID: {row[0]}, Hospital Name: {row[1]}, Bed Count: {row[2]}")
print()

query = """
CREATE TABLE IF NOT EXISTS doctor_details (
```

```

        id INT AUTO_INCREMENT PRIMARY KEY,
        doctor_name VARCHAR(255),
        hospital_id INT,
        speciality VARCHAR(255),
        salary DECIMAL(10, 2),
        experience INT
    )
    """
cursor.execute(query)
print("Table 'doctor_details' created successfully")

insert_query = "INSERT INTO doctor_details (doctor_name, hospital_id, speciality,
salary, experience) VALUES (?, ?, ?, ?, ?)"
data = [
    ("Karan", 1, "Pediatric", 40000.00, 0),
    ("Naresh", 1, "Oncology", 80000.00, 5),
    ("Hardik", 2, "Surgery", 60000.00, 2),
    ("Vishal", 2, "Homeopathy", 50000.00, 1),
    ("Jay", 3, "Ayurvedic", 40000.00, 0),
    ("Deep", 3, "Physiotherapist", 70000.00, 4),
    ("Divyesh", 4, "Pediatric", 55000.00, 3),
    ("Arjun", 4, "Scin", 55000.00, 3),
]
cursor.executemany(insert_query, data)
cnx.commit()
print(f"{cursor.rowcount} records inserted successfully")

query = "SELECT * FROM doctor_details"
cursor.execute(query)
results = cursor.fetchall()
print("\nDoctor Details:")
for row in results:
    print(f"ID: {row[0]}, Name: {row[1]}, Specialization: {row[3]}, Salary:
{row[4]}, Experience: {row[5]}")

```

Output:

Table 'hospital_details' created successfully
4 records inserted successfully

Hospital Details:

ID: None, Hospital Name: Janta, Bed Count: 200
ID: None, Hospital Name: Zydus, Bed Count: 500
ID: None, Hospital Name: Sal, Bed Count: 1000
ID: None, Hospital Name: Stirling, Bed Count: 1500

Table 'doctor_details' created successfully
8 records inserted successfully

Doctor Details:

ID: None, Name: Karan, Specialization: Pediatric, Salary: 40000,
Experience: 0
ID: None, Name: Naresh, Specialization: Oncology, Salary: 80000,
Experience: 5
ID: None, Name: Hardik, Specialization: Surgery, Salary: 60000,
Experience: 2
ID: None, Name: Vishal, Specialization: Homeopathy, Salary: 50000,
Experience: 1
ID: None, Name: Jay, Specialization: Ayurvedic, Salary: 40000,
Experience: 0
ID: None, Name: Deep, Specialization: Physiotherapist, Salary:
70000, Experience: 4
ID: None, Name: Divyesh, Specialization: Pediatric, Salary: 55000,
Experience: 3
ID: None, Name: Arjun, Specialization: Scin, Salary: 55000,
Experience: 3